

交互题

人员

李瑞涵、田心一、柳力玮、赵书梵、刘祺、刘宸熙、初锦阳 到课, 杨咏丞、高健桓、蒋叔璋 线上

上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4953>

#	用户名	姓名	编程分	时间	A (12 / 29)	B (11 / 28)	C (11 / 22)	D (11 / 16)	E (8 / 10)	F (3 / 3)	G (1 / 3)
1	liuliwei	柳力玮	7	83:48:52	0:12:37	0:19:6	(-3) 0:57:29	0:30:37 ♥	1:32:22 ♥	1:57:10 ♥	(-2) 76:39:31 ♥
2	liuqi	刘祺	6	16:5:22	(-5) 0:16:48	(-1) 0:46:10	1:20:23	1:47:35	4:50:42	5:3:44	
3	yangyongcheng	杨咏丞	6	451:50:58	(-1) 0:36:3	1:30:36	99:34:43	99:51:14	124:53:50	125:4:32	
4	tianxinyi	田心一	5	5:3:44	0:11:52	0:18:24 ♥	(-2) 0:57:32	1:21:55	1:34:1		
5	mingzongliang	明宗亮	5	6:27:4	0:11:15 ♥	0:29:0	(-3) 1:18:37	1:32:11	1:56:1		
6	zhaoshufan	赵书梵	5	6:51:52	(-2) 0:17:14	(-1) 0:50:48	(-1) 0:50:4 ♥	1:19:12	(-1) 1:54:34		
7	yuanzhao	苑钊	5	8:9:30	0:32:23	(-1) 0:48:27	1:10:16	(-3) 2:1:24	(-1) 1:57:0		
8	li Ruihan	李瑞涵	5	8:28:44	(-3) 0:31:0	(-3) 0:46:22	1:19:39	1:46:12	2:5:31		
9	jibohan	纪博涵	4	3:51:48	您正在共享屏幕 结束共享 22:36			1:34:6			UP
10	liuchenxi	刘宸熙	4	6:57:34	(-3) 0:34:14	1:35:4	1:43:53	2:4:23			

本周作业

<https://cppoj.kids123code.com/contest/5145> (课上讲了 A ~ D 题, 课后作业是 A ~ D 题必做)

课堂表现

今天给同学们讲了 交互题 的内容, 同学们课上接受的还不错, 第四题会比较难一些, 需要同学们课下好好复习复习。

课堂内容

猜数 (IO交互版)

交互题模板

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
```

```

int l = 1, r = 1e9;
while (true) {
    int mid = (l + r) / 2;
    cout << mid << endl;

    int x; cin >> x;
    if (x == 0) break;
    if (x == -1) l = mid+1;
    else r = mid-1;
}
return 0;
}

```

nextmin

题目允许错一个, 把所有位置问一遍即可

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 1000 + 5;
int w[maxn];

int main()
{
    int n; cin >> n;
    for (int i = 1; i <= n; ++i) {
        cout << "? " << i << " " << i << endl;
        int x; cin >> x;
        w[i] = x;
    }

    cout << "! ";
    for (int i = 1; i <= n; ++i) cout << w[i] << " ";
    return 0;
}

```

[COI 2020] Zagrade

括号匹配, 每次判断 i 和 栈顶 之间合不合法

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e5 + 5;
char s[maxn];

```

```

int main()
{
    int n, Q; cin >> n >> Q;
    stack<int> stk;
    for (int i = 1; i <= n; ++i) {
        if (stk.empty()) stk.push(i);
        else {
            int j = stk.top();
            cout << "? " << j << " " << i << endl;

            int x; cin >> x;
            if (x == 1) s[j] = '(', s[i] = ')', stk.pop();
            else stk.push(i);
        }
    }

    int len = stk.size();
    for (int i = 1; i <= len/2; ++i) {
        int x = stk.top(); s[x] = '('; stk.pop();
    }
    for (int i = 1; i <= len/2; ++i) {
        int x = stk.top(); s[x] = ')'; stk.pop();
    }

    cout << "! ";
    for (int i = 1; i <= n; ++i) cout << s[i];
    cout << endl;
    return 0;
}

```

[COI 2022] 回文子串 / Madioničar

枚举每个位置作为中心, 往左右枚举长度, 看对应串是不是回文

```

#include <bits/stdc++.h>

using namespace std;

int main()
{
    int n; cin >> n;

    int res = 1;
    for (int i = 1; i <= n; ++i) {
        int len = (res - 1) / 2 + 1;
        while (true) {
            int l = i - len, r = i + len;
            if (l < 1 || r > n) break;

            cout << "? " << l << " " << r << endl;
            int x; cin >> x;

```

```
        if (x == 0) break;

        ++len;
    }
    res = 2*(len-1) + 1;
}

for (int i = 1; i <= n-1; ++i) {
    int len = (res - 1) / 2 + 1;
    while (true) {
        int l = i - len + 1, r = i + len;
        if (l<1 || r>n) break;

        cout << "? " << l << " " << r << endl;
        int x; cin >> x;
        if (x == 0) break;

        ++len;
    }
    res = max(res, 2*(len-1));
}

cout << "! " << res << endl;
return 0;
}
```